

What Must the Price in Decentralized Exchanges Be?

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Abstract—Decentralized exchanges are one of the most remarkable revolutionary inventions in cryptocurrency trading. Decentralized exchanges present a way to set prices mathematically without maintaining the order book-based trading system. The order book-based trading system has been a widely-known method to determine prices in the stock markets for over 400 years. Recently, a new type of decentralized exchange has devised a constant function market maker to determine prices mathematically. Among them, the method of constant product market makers is the most widely used one. In addition, several constant function market makers have been proposed. However, there was little discussion about the desirable properties of a constant function market maker regarding price and impermanent loss. In this paper, we discuss the desirable price and its effects on impermanent loss. This study considers two types of prices: the reference price and the actual price. We show that impermanent gain can be achievable under certain conditions and prove it mathematically. Two examples are provided to show that impermanent gain is achievable. One question is whether there is agreement on applying actual price when calculating value gains.

Keywords—*blockchain, cryptocurrency, decentralized exchange, impermanent loss, market maker, smart contract.*

I. INTRODUCTION

- Order book-based trading systems have a history of more than 400 years in the stock market. The first generation of centralized cryptocurrency exchanges have also adopted order book-based trading systems of stock exchanges. It seems to be because everyone was accustomed to the order book trading system. Early decentralized exchanges such as EtherDelta or 0x over Ethereum mainnet also adopted order book system. When the price of a stock is unknown, an auction is commonly used to determine the price. The order book trading system relies on a double auction or continuous double auction (CDA) method as a price discovery mechanism. The sellers and buyers of a certain stock propose desired prices and quantities to the other party, respectively. If the price and quantity match, the transaction is made. In a CDA, buyers and sellers can continuously

update offers and bids. A CDA-based trading system maintains two priorities: Orders are matched based on price priority (better prices are executed first) and time priority (earlier orders are executed first if prices are the same).

Maintaining an order book through a smart contract on the Ethereum blockchain is costly and tends to be slow due to the complicated matching mechanism of an order book system, and the limited memory capacity of the blockchain [1]. As a result, an off-chain order book solution has been introduced. The two-layer decentralized trading system reduces costs and increases speed by allocating trader matching activities off-chain and trade settlement activities on-chain. Once trades are matched in the off-chain order book, settlement takes place on-chain. Later, newly emerged decentralized exchanges such as Bancor and Uniswap over Ethereum, Raydium over Solana, and Quickswap over Polygon started to use automated market makers (AMMs), which do not require order books for trading.

While most AMM-based exchanges rely solely on liquidity pools, some hybrid platforms like Serum (with Raydium) and Loopring combine AMMs with order books to create a more flexible and efficient trading experience. These hybrid models are becoming increasingly popular as they address the several limitations of pure AMMs, such as slippage and impermanent loss, while retaining the benefits of decentralized trading.

AMM without order book is a remarkable innovation since there is no price discovery process. AMM decides the price using a mathematical function, regardless of what buyers and sellers want. Traders can decide only when and how much to trade. However, in reality, it is difficult to maintain time-priority and price-priority in a decentralized exchange. This is because the sales order and purchase order transactions stored in the mempool go through a process of selection by validators. In this process, the maximal extractable value (MEV) comes into play.

In the past, mathematical pricing methods did not exist, but now, beyond expectations, they are widely used in the market. AMM's smart contract determines the price immediately with no exceptions according to pre-defined rules and it leads to a sense of fairness in trading. In the decentralized exchanges with

AMMs, there are traders and liquidity providers (LPs) who facilitate effective trading. Impermanent loss is an inherent feature of AMMs due to their design and reliance on constant product formulas. Impermanent loss is the temporary loss of funds that liquidity providers may experience when providing liquidity to an AMM pool due to price changes in the assets. It is a key consideration for LPs in decentralized finance, and understanding it is crucial for managing risk and maximizing returns. Impermanent loss is one of the most significant research topics in decentralized finance [2][3][4][5][6][7][8]. Among those, most studies were devoted to impermanent loss in CPMM (constant product market maker). It is important to note that almost half of liquidity providers are losing money due to impermanent loss [9].

Bancor [10] was the first decentralized exchange that introduced a liquidity pool smart contract on the Ethereum network in 2017 [11]. The liquidity pool smart contract allows traders to buy and sell assets immediately at any time since each liquidity pool consists of a pair of base and pricing assets deposited. For example, Ethereum (ETH) is the base asset to be traded, and USDT or USDC or whatever is the pricing asset (a.k.a. stablecoin) in the liquidity pool. The price of the base asset is set mathematically by the smart contract. Thus, asset trading can be completed in an instant due to the abundant liquidity and mathematical pricing.

A few years before Bancor, Hanson [12][13] introduced LMSR (logarithmic market scoring rule) for the prediction market. The LMSR can be used as an automated market maker [14]. Othman *et al.* [15] proposed LS-LMSR (liquidity-sensitive logarithmic market scoring rule), which is an advanced version of LMSR. However, the LMSR-type AMMs require intensive computations to determine prices through smart contracts and have several other deficiencies. Several decentralized prediction markets, including Augur and Gnosis used the LMSR-type AMMs. However, Augur and Gnosis are prediction market platforms and not decentralized exchanges.

Uniswap [16] has adopted CPMM (constant product market maker) as an AMM and showed that it is sufficiently popular, effective and sustainable, growing as the leading decentralized exchange. Later, Balancer [17] introduced the CMMM (constant mean market maker), which is a variant of the CPMM with less impermanent loss. In Wang's study [14], CSMM (constant sum market maker) and CCMM (constant circle market maker) are introduced. Recently, Angeris and Chitra [17] generalized existing AMMs as CFMM (constant function market maker). Now, many decentralized exchanges are using CPMM since it is mathematically simple to implement and consumes less gas than the LMSR-type AMMs.

Impermanent loss can be considered as a penalty imposed on liquidity providers in all AMMs. It occurs in a situation where liquidity providers gain more benefit from not depositing assets than depositing them into the liquidity pool. However, most liquidity providers still provide liquidity despite the loss because they expect transaction fees to compensate for the loss and to obtain governance token rewards. In addition, most AMMs have failed to avoid impermanent loss. Recently, Kim *et al.* [6] computed the impermanent loss of four automated market makers under the assumption that the price of an asset in a

liquidity pool is the ratio of the numbers of two assets and showed that impermanent gain is achievable in certain cases. However, this case is limited effectively to only the CPMM. As Kim *et al.* [21] showed, the only case where the price is expressed as a ratio of the numbers of two assets is the CPMM of Uniswap v1. Recently, Lee and Kim [22] showed that CPMM impermanent loss can be mitigated by utilizing batching and reordering of multiple transactions. To understand the basic concept of the impermanent loss mitigation scheme, note that the impermanent loss is computed using the price after trading is completed, with the number of initial assets deposited into the liquidity pool and the number of assets after trading.

In this paper, we review two types of price functions, which are important in determining impermanent loss. The first price is the reference price, which is roughly calculated before the actual asset trading. The second price is the actual price, which is accurately calculated after the actual trading. The difference between the two prices is called slippage. Slippage occurs when traders use the reference price, but not when they use the actual price. This is because the former is the expected price just before the transaction, while the latter is the actual transaction price. Distinguishing between these two prices is one of the important contributions of this paper. This is because the impermanent loss curve was drawn using the reference price. Unfortunately, by using this reference price, impermanent loss seems unavoidable. Kim *et al.* [22] have delved into the properties of the price in the various CFMMs.

If we draw the impermanent loss curve using the actual price, we can get an unexpected result. It has been known that impermanent loss is unavoidable, always occurs, and does not occur only when returning to the initial state. However, if we draw the impermanent loss curve using the actual price, we can see that under certain conditions, gains are possible instead of losses. The second contribution of this paper is that, unlike previous research results that impermanent loss is unavoidable, we show that impermanent gain is also possible. Needless to say, this paper explains under what conditions impermanent gain occurs. In addition, this paper emphasizes the importance of this discovery by comparing with previous research results. This paper also provides a background of the theoretical result of the optimized impermanent loss mitigation method [23].

II. MATHEMATICAL PRICE AND VALUE GAIN

In this paper, we consider only a constant product market maker cost function containing two variables x and y . The constant cost function such as $C(\mathbf{q}) = k$ is called constant function market maker, where k is a constant. A state $\mathbf{q}$ has two variables x and y , representing the numbers of two tokens under consideration, respectively.

The price p_x of x and the price p_y of y are obtained by differentiating the cost function as follows.

$$p_x = \frac{dC(\mathbf{q})}{dx} \quad \text{and} \quad p_y = \frac{dC(\mathbf{q})}{dy}.$$

When the sum of the two prices is 1 (i.e. $p_x + p_y = 1$), the price is defined as translation invariant. Therefore, since the price of the stablecoin is already set at 1, it is unreasonable to use this

price in the exchange. However, translation invariance is a very important condition in prediction markets.

Automatic market makers of decentralized exchanges use a different pricing system. As the price of the stablecoin is set at 1, so P_y is always 1 because the stablecoin is used as a unit of account. $P_x(\mathbf{q}_t - \mathbf{q}_{t-1})$, the actual price, is determined as the ratio of the difference between x_{t-1} and x_t and the difference between y_{t-1} and y_t . In other words,

$$P_x(\mathbf{q}_t - \mathbf{q}_{t-1}) = -\frac{y_t - y_{t-1}}{x_t - x_{t-1}}.$$

If the difference between x_{t-1} and x_t is infinitesimal, we obtain

$$P_x(\mathbf{q}_t) = \lim_{x_t \rightarrow x_{t-1}} -\frac{y_t - y_{t-1}}{x_t - x_{t-1}} = -\frac{dy_t}{dx_t} = \frac{p_x}{p_y}.$$

The value of the assets in the automated market makers is given as a product of asset price and number as follows:

$$V_x = P_x \cdot x \text{ and } V_y = y.$$

Impermanent loss is a terminology created on the premise that there will be no time when the current value of the liquidity provider's assets is more profitable than before. However, it misleads us to a belief that permanent or impermanent gain can never exist. Thus, this paper introduces a neutral definition of liquidity provider's value gain, G , as follows:

$$G_t = P_x \cdot (x_t - x_0) + (y_t - y_0).$$

where x_0 and y_0 refer to the quantities of the base and the pricing assets at the time the liquidity provider first entrusted the asset pair, respectively. If $G_t \leq 0$, then the liquidity provider incurs impermanent loss.

We refer to $P_x(\mathbf{q}_t - \mathbf{q}_{t-1})$ and $P_x(\mathbf{q}_t)$ as the actual price and the reference price, respectively. It is trivial that these two prices are not equal. Note that the actual price is a first-order Markov process, and the reference price is a zero-order Markov process. A zero-order Markov process is memoryless, with future states independent of past and current states. In a first-order Markov process, future state depends only on the current state. The curve of value gain can be drawn using

$$G_t(\mathbf{q}_t) = P_x(\mathbf{q}_t) \cdot (x_t - x_0) + (y_t - y_0).$$

In this case, it is clear that

$$G_t(\mathbf{q}_t) \leq 0.$$

That is, impermanent loss occurs at all times. This is what has been known so far. However, there is another value gain computed using the actual price as follows:

$$G_t(\mathbf{q}_t - \mathbf{q}_{t-1}) = P_x(\mathbf{q}_t - \mathbf{q}_{t-1}) \cdot (x_t - x_0) + (y_t - y_0).$$

We show that this value gain can lead to impermanent gain, which is different from what has been known.

III. CPMM

The CPMM cost function, which is used in Uniswap [1], is a product of x_t and y_t as follows:

$$C(\mathbf{q}_t) = x_t \cdot y_t = k.$$

Then,

$$p_x(\mathbf{q}_t) = \frac{dC(\mathbf{q}_t)}{dx_t} = y_t, \quad p_y(\mathbf{q}_t) = \frac{dC(\mathbf{q}_t)}{dy_t} = x_t.$$

Thus,

$$P_x(\mathbf{q}_t) = \frac{p_x(\mathbf{q}_t)}{p_y(\mathbf{q}_t)} = \frac{y_t}{x_t} = \frac{k}{x_t^2}.$$

As explained above, if this reference price is used to compute value gain $G_t(\mathbf{q}_t)$, it cannot be larger than 0, i.e., impermanent loss occurs.

We show the results of value gain $G_t(\mathbf{q}_t - \mathbf{q}_{t-1})$, if the actual price is used. To obtain an insight, we start with a simple example. Assume both initial values of x_0 and y_0 be 10. In this case, the constant k is 100. In order for the constant k to remain 100 when x_{t-1} is 5, y_{t-1} must be 20 at time $t - 1$. Assume that a transaction has been made. Now, if x_t is 20, y_t must be 5 to ensure that the constant k remains 10. It is natural that someone put 15 tokens of x into and take out 15 tokens of y from the liquidity pool. In this situation, the actual price of token x is calculated as follows:

$$P_x(\mathbf{q}_t - \mathbf{q}_{t-1}) = -\frac{5 - 20}{20 - 5} = 1.$$

Using this actual price of 1 to calculate the value gain, we obtain the following result:

$$G_t(\mathbf{q}_t - \mathbf{q}_{t-1}) = 1 \cdot (20 - 10) + (5 - 10) = 5.$$

Surprisingly, the value gain is positive at 5. No loss occurs, and a profit is made.

In the next example, assume x_0 is 10 and y_0 is 10. Let x_{t-1} and y_{t-1} be 20 and 5, respectively, at time $t - 1$. At the next transaction, x_t becomes 5, y_t becomes 20. In this situation, the actual price is computed as follows:

$$P_x(\mathbf{q}_t - \mathbf{q}_{t-1}) = -\frac{20 - 5}{5 - 20} = 1.$$

Using this actual price of 1 to calculate the value gain, we still obtain the following result:

$$G_t(\mathbf{q}_t - \mathbf{q}_{t-1}) = 1 \cdot (5 - 10) + (20 - 10) = 5.$$

Surprisingly, the value gain is positive again at 5. Similarly, no loss occurs, and a profit is made, too.

It is an important discovery which is straightforward and obvious. We conjecture that the condition for impermanent gain to occur is as follows: " $x_{t-1} < x_0 < x_t$ " or " $x_t < x_0 < x_{t-1}$ ".

Theorem 1. Assume that $x_t \cdot y_t = k$ for all non-negative t with a positive constant k (the case of CPMM). If x_{t-1} is x_0 , then value gain $G_t(\mathbf{q}_t - \mathbf{q}_0)$ using the actual price is zero.

Proof. Since $x_{t-1} = x_0$, it is clear that $y_{t-1} = y_0$ from the formula $x_t \cdot y_t = k$. Thus,

$$P_x(\mathbf{q}_t - \mathbf{q}_0) = -\frac{y_t - y_{t-1}}{x_t - x_{t-1}} = -\frac{y_t - y_0}{x_t - x_0}$$

and

Denote point (x_0, y_0) as the origin of our study. Theorem 1 proves that the quantity changes from the origin to another point on the hyperbola always makes the value gain zero; that is, it does not cause either impermanent loss or gain. Thus, using Theorem 1, we can determine the boundary condition between impermanent loss and impermanent gain.

Theorem 2. Assume that $x_t \cdot y_t = k$ for all non-negative t with a positive constant k (the case of CPMM). If $x_{t-1} < x_0 < x_t$, then value gain $G_t(\mathbf{q}_t - \mathbf{q}_{t-1})$ using the actual price is always positive.

Proof. Since $x_{t-1} < x_0 < x_t$, is clear that $y_{t-1} > y_0 > y_t$ from the formula $x_t \cdot y_t = k$. Figure 1 shows three points A , B , and D , on the hyperbolic curve $x_t \cdot y_t = k$. The coordinates of vertices A , B , and D are (x_{t-1}, y_{t-1}) , (x_t, y_t) , and (x_0, y_0) , respectively. $P_x(\mathbf{q}_t - \mathbf{q}_{t-1})$ and $P_x(\mathbf{q}_t - \mathbf{q}_0)$ are defined as follows:

$$P_x(\mathbf{q}_t - \mathbf{q}_{t-1}) = -\frac{y_t - y_{t-1}}{x_t - x_{t-1}},$$

and

$$P_x(\mathbf{q}_t - \mathbf{q}_0) = -\frac{y_t - y_0}{x_t - x_0}.$$

In Figure 1, extending line segment DB creates line segment EB. Then, triangles DBE and FBC are congruent. Therefore,

$$P_x(\mathbf{q}_t - \mathbf{q}_0) = -\frac{y_t - y_0}{x_t - x_0} = -\frac{y_t - y'_{t-1}}{x_t - x_{t-1}}.$$

In the above price formulas $P_x(\mathbf{q}_t - \mathbf{q}_{t-1})$ and $P_x(\mathbf{q}_t - \mathbf{q}_0)$, the denominators are the same but the numerators are different. Since $(y'_t - y_{t-1}) < (y_t - y_{t-1}) < 0$, it is clear that $(y_{t-1} - y'_t) > (y_{t-1} - y_t) > 0$. Since $(x_t - x_{t-1}) > 0$, therefore,

$$P_x(\mathbf{q}_t - \mathbf{q}_{t-1}) > P_x(\mathbf{q}_t - \mathbf{q}_0)$$

or

$$G_t(\mathbf{q}_t - \mathbf{q}_{t-1}) = -\frac{y_t - y_{t-1}}{x_t - x_{t-1}}(x_t - x_0) + (y_t - y_0) > 0.$$

This concludes the proof. $\square$

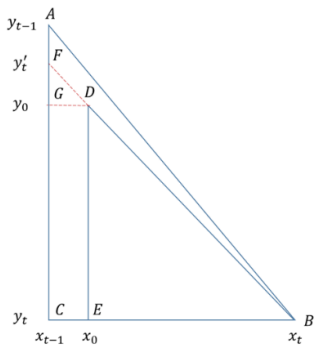


Figure 1. Three points on a hyperbolic curve $x_t \cdot y_t = k$, where $x_t > x_0 > x_{t-1}$ and $y_t < y_0 < y'_t < y_{t-1}$.

Theorem 3. Assume that $x_t \cdot y_t = k$ for all non-negative t with a positive constant k (the case of CPMM). If $x_t < x_0 < x_{t-1}$, then value gain $G_t(\mathbf{q}_t - \mathbf{q}_{t-1})$ using the actual price is always positive.

Proof. Since $x_t < x_0 < x_{t-1}$, is clear that $y_t > y_0 > y_{t-1}$ from the formula $x_t \cdot y_t = k$. Figure 2 shows three points A , B , and D , on the hyperbolic curve $x_t \cdot y_t = k$. The coordinates of vertices A , B , and D are (x_t, y_t) , (x_{t-1}, y_{t-1}) , and (x_0, y_0) , respectively. $P_x(\mathbf{q}_t - \mathbf{q}_{t-1})$ and $P_x(\mathbf{q}_t - \mathbf{q}_0)$ are defined as follows:

$$P_x(\mathbf{q}_t - \mathbf{q}_{t-1}) = -\frac{y_t - y_{t-1}}{x_t - x_{t-1}},$$

and

$$P_x(\mathbf{q}_t - \mathbf{q}_0) = -\frac{y_t - y_0}{x_t - x_0}.$$

If we extend line segment AD, we get line segment AE. Therefore, triangles ADG and AEC are congruent (see Figure 2). When we draw line segment EF parallel to line segment AB, triangles ABC and FEC are also congruent.

From Figure 2 and using the congruence of triangles,

$$P_x(\mathbf{q}_t - \mathbf{q}_{t-1}) = -\frac{y_t - y_{t-1}}{x_t - x_{t-1}} = -\frac{y'_t - y_{t-1}}{x_t - x'_0},$$

and,

$$P_x(\mathbf{q}_t - \mathbf{q}_0) = -\frac{y_t - y_0}{x_t - x_0} = -\frac{y_t - y_{t-1}}{x_t - x'_0}.$$

In the above price formulas $P_x(\mathbf{q}_t - \mathbf{q}_{t-1})$ and $P_x(\mathbf{q}_t - \mathbf{q}_0)$, the denominators are the same but the numerators are different. Since $(y'_t - y_{t-1}) < (y_t - y_{t-1})$, but $(x_t - x'_0) < 0$, it is clear that

$$P_x(\mathbf{q}_t - \mathbf{q}_{t-1}) > P_x(\mathbf{q}_t - \mathbf{q}_0)$$

or

$$G_t(\mathbf{q}_t - \mathbf{q}_{t-1}) = -\frac{y_t - y_{t-1}}{x_t - x_{t-1}}(x_t - x_0) + (y_t - y_0) > 0.$$

This concludes the proof. $\square$

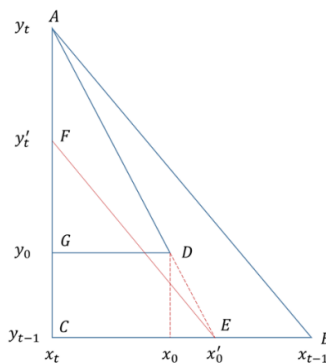


Figure 2. Three points on a hyperbolic curve $x_t \cdot y_t = k$, where $x_t < x_0 < x'_t < x_{t-1}$ and $y_t > y'_t > y_0 > y_{t-1}$.

Theorem 2 and Theorem 3 show the necessary condition of impermanent gain, which is “ $x_t < x_0 < x_{t-1}$ ” or “ $x_t > x_0 > x_{t-1}$ ” and using the actual price.

Figure 3 shows a new curve of value gain, $G_t(\mathbf{q}_t - \mathbf{q}_{t-1})$ with the actual price. The blue curve is the value gain curve of $G_t(\mathbf{q}_t)$ with the reference price. The x -axis represents the value of x_t , and the y -axis represents the value gain. If the value gain is less than 0, it represents a loss, and if a value gain is larger than 0, it represents a gain. The blue curve shows a typical impermanent loss curve, which is 0 or below 0.

In Figure 3, Let $x_0 = 10$ and $y_0 = 10$, so that $k = 100$. When $x_{t-1} = 14$, follow the red curve from $x_t = 6$ to $x_t = 19$. A clear impermanent gain can be observed on the left-hand side of the origin, i.e., $x_0 = 10$, and impermanent loss on the right-hand side. Two curves intersect twice at $x_0 = 10$ and $x_{t-1} = 14$.

IV. CONCENTRATED LIQUIDITY CPMM

Uniswap V3 uses concentrated liquidity, such as

$$\left(x_t + \frac{k}{\sqrt{P_b}}\right)(y_t + k\sqrt{P_a}) = k^2.$$

This CPMM is effective for

$$0 \leq x_t \leq k \frac{\sqrt{P_b} - \sqrt{P_a}}{\sqrt{P_a} \cdot \sqrt{P_b}}.$$

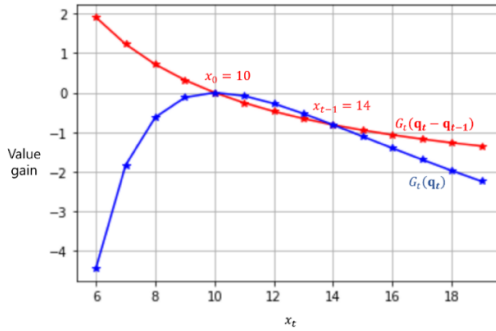


Figure 3. A value gain curves of CPMM.

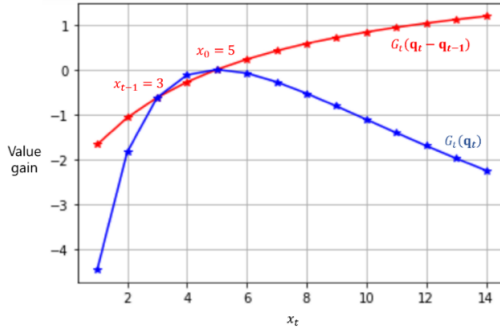


Figure 4. A value gain curves of concentrated liquidity CPMM.

Its price range is limited to

$$P_a \leq P_x(\mathbf{q}_t) \leq P_b.$$

The price of the Uniswap V3 market maker is represented as follows:

$$P_x(\mathbf{q}_t) = \frac{k^2}{\left(x_t + \frac{k}{\sqrt{P_b}}\right)^2}$$

Given x_t, y_t can be computed as follows:

$$y_t = \frac{k^2}{\left(x_t + \frac{k}{\sqrt{P_b}}\right)^2} - k\sqrt{P_a}.$$

We want to show an example in Figure 4. For constant $k = 10, P_a = 1/4, P_b = 4, x_0 = 5$, and $y_0 = 5$, we can compute the value gain using $P_x(\mathbf{q}_t)$ and $P_x(\mathbf{q}_t - \mathbf{q}_{t-1})$. The red curve in Figure 4 is the value gain curve calculated using the actual price. It shows that impermanent gain occurs when it surpasses the origin. The origin x_0 is 5 and x_{t-1} is 3. If x_{t-1} is 3 and x_t is less than 5, a loss occurs. However, if x_{t-1} is 3 and x_t is greater than 5, a gain occurs. The blue curve is the value gain using the reference price. This blue curve shows that impermanent loss occurs at all times. Comparing Figures 3 and 4, we can see that CPMM and concentrated liquidity CPMM behave similarly. This is because both methods are part of the CPMM family. It is known that concentrated liquidity CPMM is a piece-wise CPMM.

V. PRICE FUNCTIONS

It was not easy to set prices for anyone. That is why so many auction systems were introduced. If mathematical formulas could determine prices in the market, there would not have been much use for the auction system. The reason why a seller uses calculators before telling customers the price is to do simple additions or multiplications. The purpose may be to assure customers by giving them the impression that the seller has calculated accurately. The seller does not intend to use formulas to determine prices. Moreover there were no mathematical formulas for clerks to use. However, it is historical that AMM consistently used mathematical formulas to determine prices without exception. Even though the prices were determined by a mathematical formula, it is surprising that customers accepted the prices determined in that way. It implies that customers recognized that pricing based on mathematical formulas was reasonable. CPMM reflected the principle that prices decrease when quantity increases and increase when quantity decreases, so it must have seemed reasonable.

The reference price is easy to calculate and mathematically attractive, but it is not a real price. In particular, there is one more principle that must be followed in CPMM in addition to the principle of supply and demand. That is the principle that constant k must be the same before and after the transactions. In order to not change constant k after the transactions, the CPMM formula is used to determine how many assets must exist in the liquidity pool. The CPMM formula is a tool to calculate the target number of assets after the transaction. However, people

have been determining the reference price using the target number of assets. They have also been calculating the value gain using it. So, they did not realize that impermanent gains can occur. What is clear is that impermanent gains never occur when calculating the value gain using the reference price. Impermanent gains can occur when calculating the value gain using the actual price. It is true that impermanent gains do not always occur. Among tens of thousands of transactions, gains may only occur a few dozen times. The important thing is that since gains can occur, they can be used well to encourage liquidity providers and activate AMMs. Lee and Kim [23] has tried to exploit this fact to maximize the gain.

Some block producers maximize their profits by using the MEV [24]. MEV refers to the strategy of including, omitting, or reordering transactions when creating a new block. The goal of MEV is to generate as much additional profit as possible. Using MEV can maximize the profits of liquidity providers. However, MEV can be treated as a crime [25]. Therefore, a method has been proposed to provide profits to liquidity providers as if they were using MEV without using MEV [23].

What we call “price” is neither the future price that will be realized nor the price that has not yet been realized but the price that has already been realized. The price that Picasso's painting was traded at the end is the actual price. The price that the owner of the painting wants, i.e., the future price, is not the actual price. Similarly, the reference price is the one that has not been realized and is different from the actual price. There are many reasons why the actual price should be the official price. The actual price is the increment of x versus the increment of y . To calculate the increment, we need to know x_t and x_{t-1} . Thus, the actual price is similar to a first-order Markov process. Since the reference price determines the price only with x_t without considering x_{t-1} , it is similar to a zero-order Markov process. The debate over whether to use reference prices, actual prices, or external prices will likely have to continue in the future.

VI. CONCLUSION

In this paper, we investigated if impermanent loss can be avoided when applying actual price in CPMM and concentrated liquidity CPMM. We showed impermanent gain is achieved when the actual price is used under a condition $x_t < x_0 < x_{t-1}$ or $x_t > x_0 > x_{t-1}$. The question is whether there is agreement on applying actual price when calculating value gains.

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